How-To Library

Informatica

How to Create Ephemeral Clusters with Data Engineering 10.4.1

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Abstract

You can use a workflow to automate the creation of a cluster on supported cloud platforms. The workflow creates a cluster and runs mappings and other workflow tasks. When you include a Delete Cluster task, so that the cluster terminates when workflow tasks are complete, the cluster is known as an ephemeral cluster.

Supported Versions

- Informatica Data Engineering Integration 10.4.1
- Informatica Enterprise Data Catalog 10.4.1
- Informatica Enterprise Data Preparation 10.4.1

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Cluster Workflows Overview

A cluster workflow is a workflow that automates the creation of a cluster using a Create Cluster task, and then runs one or more mappings on that cluster. You can create a cluster workflow to run Mapping and other tasks on a cloud platform cluster. To use an ephemeral cluster to run mappings in the workflow, include a Delete Cluster task to terminate and delete the cluster after the mappings run.

The cluster workflow uses other elements that enable communication between the Data Integration Service and the cloud platform, such as a cloud provisioning configuration and a cluster connection.

A cluster workflow contains a Create Cluster task that you configure with information about the cluster to create. If you want to create an ephemeral cluster, you can include a Delete Cluster task. An ephemeral cluster is a cloud platform cluster that you create to run mappings and other tasks, and then terminate when tasks are complete. Create ephemeral clusters to save cloud platform resources.

You can use cluster workflows to create Hadoop clusters on the Microsoft Azure or Amazon AWS cloud platforms in a Hadoop environment, or Databricks clusters in a Databricks environment on Azure or AWS.

Cluster Workflows Platform Support

Cluster workflow capabilities vary depending on the environment and the cloud platform.

Hadoop Clusters

On the Azure platform, you can create an ephemeral HDInsight cluster that accesses ADLS Gen2 resources.

On the AWS platform, you can create an ephemeral Amazon EMR cluster to access S3, Redshift, and Snowflake resources.

On Cloudera Altus, you create a workflow with Command tasks that perform the tasks that a cluster workflow automates. For more information about creating a cluster on Cloudera Altus, see the article "How to Create Cloudera Altus Clusters with a Cluster Workflow" on the Informatica Documentation Portal.

Databricks Clusters

On the Azure and AWS Databricks platforms, you can configure Databricks ephemeral clusters.

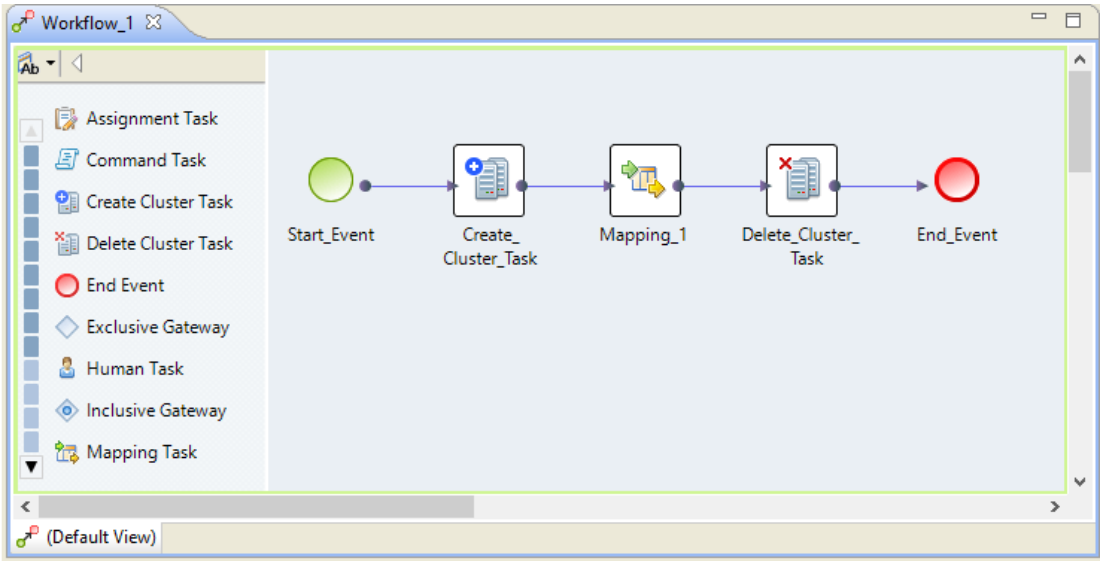
Optionally, you can configure the clusters to start from sets of cached VM instances, called warm pools. These instances wait on standby in a running state for ephemeral cluster creation. You can choose to have the instances remain on standby when the ephemeral clusters are terminated. For more information, see the Databricks documentation.

Enable warm pools for ephemeral Databricks clusters in the advanced properties of the Create Cluster workflow task.

Cluster Workflow Components

The cluster workflow that creates an ephemeral cluster includes a Create Cluster task, at least one Mapping task, and a Delete Cluster task.

The following image shows a sample cluster workflow:



Create Cluster Task

The Create Cluster task contains all the settings that Amazon EMR or Azure HDInsight requires to create a cluster with a master node and worker nodes. It also contains a reference to a Hadoop connection.

Create one Create Cluster task for a cluster workflow.

When you create a cluster workflow, you drag a Create Cluster task into the workflow editor, and then configure task properties.

Cloud Provisioning Configuration

The cloud provisioning configuration associates the Create Cluster task with the Hadoop connection associated with the workflow.

Configure the cloud provisioning configuration with information about the cloud platform account. To configure the cloud provisioning configuration, use the Administrator tool.

The Create Cluster task must include a reference to the cloud provisioning configuration.

For more information, see the *Data Engineering Administrator Guide*.

Hadoop Connection

Create a dedicated Hadoop connection to use with the cluster.

The Hadoop connection saves property values for the Data Integration Service to use for cluster workflow operations. When you run a cluster workflow, the Data Integration Service creates temporary Hadoop connections based on these values.

Mapping and Other Workflow Tasks

After the Create Cluster task, add other workflow tasks to run on the cluster.

Add one or more Mapping tasks to the workflow. You can also include Command and other workflow tasks.

Before you add mappings to Mapping tasks, you prepare the mappings. Configure the Hadoop Connection property in each mapping to designate whether to run the mapping on the cluster that the workflow creates, or on another cluster.

Delete Cluster Task

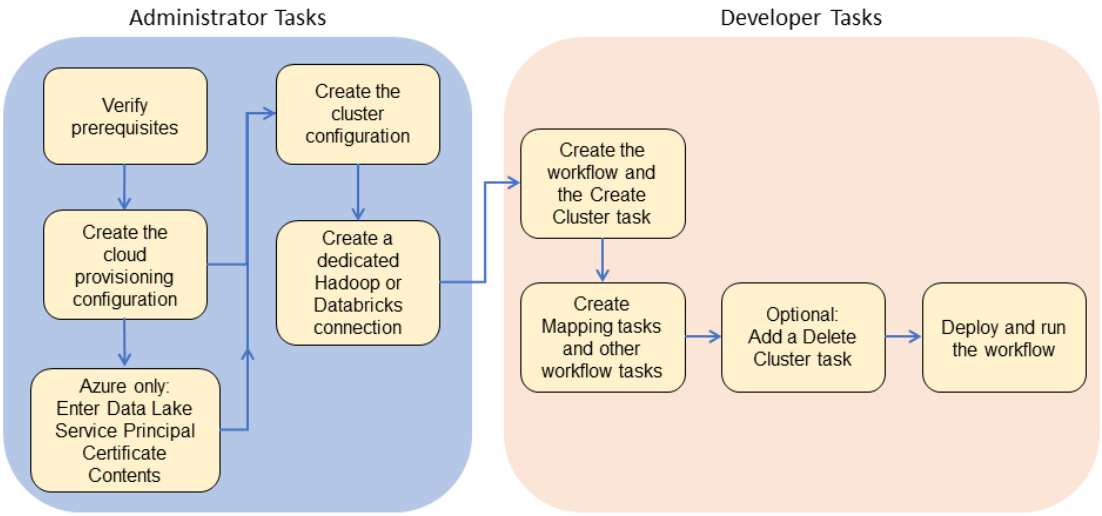
The Delete Cluster task terminates the cluster, and deletes the cluster and other resources that the cluster workflow creates.

To create an ephemeral cluster, add a Delete Cluster task to the workflow. When you do not include it, the cluster continues to run until you terminate it. You can use infacmd ccps deleteCluster to terminate the cluster at any time.

Cluster Workflows Process

Creation of a cluster workflow requires administrator and developer tasks.

The following image shows the process to create, configure, and run a cluster workflow:



The cluster workflow development process requires tasks from the following users:

Administrator

An administrator completes the following tasks:

1. Verify prerequisites for the domain and the cloud platform.
2. Create the cloud provisioning configuration in the Administrator tool.
3. Create a cluster configuration in the Administrator tool.
4. Create a dedicated cluster connection to associate with the cluster configuration.

For information about administrator tasks, see the *Data Engineering Administrator Guide*.

Developer

A developer completes the following tasks:

1. Create the workflow and the Create Cluster task.
2. Create Mapping tasks and other tasks as required.
3. Create a Delete Cluster task if you want to delete the cluster and resources when processing is complete.
4. Deploy and run the workflow.

Administrator Tasks

Before a developer creates a cluster workflow, the Informatica administrator performs the tasks in this section.

Step 1. Verify prerequisites.

Verify prerequisites for the cloud platform and the domain.

Step 2. Create a cloud provisioning configuration.

Configure the cloud provisioning configuration with information about the cloud platform account.

Step 3. Create a cluster configuration.

Import cluster information into the domain. When you import cluster information, you import values from *-site.xml files to create a domain object called a cluster configuration.

Step 4. Create a dedicated cluster connection.

Create a dedicated connection to use with the cluster workflow. When you run a cluster workflow, the Data Integration Service uses settings in the connection to run jobs on the cluster.

Verify Prerequisites

Before you create a cluster configuration, verify cloud platform and domain prerequisites.

Verify the following prerequisites for the Informatica domain:

- An Informatica domain must be integrated with the cloud platform where you plan to run ephemeral clusters. The domain can reside on an instance on the AWS or Microsoft Azure cloud platform, or on an on-premises machine. If the Informatica instance is installed on-premises, you must configure the VPN to connect to the AWS VPC or Azure vnet where the cluster runs.
- You must have permissions to create connections on the domain.
- To create clusters on AWS, the AWS administrator must open the required ports for the security group to use in the VPC where you want to create the cluster.

Verify the following prerequisites for the cloud platform:

- Create a user account with administrator permissions on the cloud platform.
- Create a resource on the cloud platform where you can create clusters.
  - On AWS, create a Virtual Private Cloud (VPC)
  - On Azure, create a Virtual Network (vnet)

  Note: The Informatica domain and the cluster that the workflow creates must be located on this resource.
- If the Informatica domain is installed on-premises, enable DNS resolution.

Enable DNS Resolution from an On-Premises Informatica Domain

If the Informatica domain is installed on-premises, you must enable DNS resolution over the VPN that connects the domain and the cloud platform.

You can enable DNS resolution on Amazon AWS or Microsoft Azure.

Amazon AWS

To run mappings from an on-premises deployment of Data Engineering Integration to a cluster on AWS, you must install and configure Unbound on an EC2 instance. Unbound enables DNS resolution from an on-premises network. To read how to install and configure Unbound in the AWS VPC, see the AWS documentation.

Microsoft Azure

To run mappings from an on-premises deployment of Data Engineering Integration to a cluster on Azure, you must use the Bind utility on the Azure virtural network.

Follow the steps in the Microsoft Azure article "DNS Configuration."

The article gives an example of the contents of the `/etc/bind/named.conf.options` file. You can put a list of available IP addresses on the domain network in the goodclients portion of the file. The following excerpt shows an example:

```
//Add the IP range of the joined network to this list
acl goodclients {
    1.2.3.0/24; # IP address range of the virtual network
    1.2.4.0/24;
    1.2.5.0/24;
    1.2.6.0/24;
    1.2.3.253;
    1.2.3.254;
    localhost;
    localnets;
};
```

Create the Cloud Provisioning Configuration

Create the cloud provisioning configuration and configure it with information that the domain needs to access and create resources on the cloud platform.

The properties to configure in the cloud provisioning configuration depend on the cloud platform.

1. From the **Connections** tab, right-click the Domain node and select **New** > **Connection**.

   The **New Connection** dialog box opens.
2. Choose one of the following cloud provisioning configuration types:
   - AWS Cloud Provisioning Configuration. For AWS cloud provisioning properties and values, see "AWS Cloud Provisioning Configuration Properties" on page 8.
   - Azure Cloud Provisioning Configuration. For Azure cloud provisioning properties and values, see "Azure Cloud Provisioning Configuration Properties" on page 9.
   - Databricks Cloud Provisioning Configuration. For Databricks cloud provisioning properties and values, see "Databricks Cloud Provisioning Configuration Properties" on page 12.
3. Enter property values in the configuration wizard, then click **Finish** to create the cloud provisioning configuration.

The cloud provisioning configuration appears in the list of connections in the **Domain Navigator**.

If you want to use ADLS storage with an Azure HDInsight cluster, you must run infacmd ccps updateADLSCertificate to populate the Data Lake Service Principal Certificate Contents property after you create the Azure cloud provisioning configuration. See "Complete the Azure Cloud Provisioning Configuration" on page 13.

AWS Cloud Provisioning Configuration Properties

The properties in the AWS cloud provisioning configuration enable the Data Integration Service to contact and create resources on the AWS cloud platform.

General Properties

The following table describes cloud provisioning configuration general properties:

Property	Description
Name	Name of the cloud provisioning configuration.
ID	ID of the cloud provisioning configuration. Default: Same as the cloud provisioning configuration name.
Description.	Optional. Description of the cloud provisioning configuration.
AWS Access Key ID	Optional. ID of the AWS access key, which AWS uses to control REST or HTTP query protocol requests to AWS service APIs. If you do not specify a value, Informatica attempts to follow the Default Credential Provider Chain.
AWS Secret Access Key	Secret component of the AWS access key. Required if you specify the AWS Access Key ID.
Region	Region in which to create the cluster. This must be the region in which the VPC is running. Use AWS region values. For a list of acceptable values, see AWS documentation. **Note:** The region where you want to create the cluster can be different from the region in which the Informatica domain is installed.

Permissions

The following table describes cloud provisioning configuration permissions properties:

Property	Description
EMR Role	Name of the service role for the EMR cluster that you create. The role must have sufficient permissions to create a cluster, access S3 resources, and run jobs on the cluster. When the AWS administrator creates this role, they select the “EMR” role. This contains the default AmazonElasticMapReduceRole policy. You can edit the services in this policy.
EC2 Instance Profile	Name of the EC2 instance profile role that controls permissions on processes that run on the cluster. When the AWS administrator creates this role, they select the “EMR Role for EC2” role. This includes S3 access by default.
Auto Scaling Role	Required if you configure auto-scaling for the EMR cluster. This role is created when the AWS administrator configures auto-scaling on any cluster in the VPC. Default: When you leave this field blank, it is equivalent to setting the Auto Scaling role to “Proceed without role” when the AWS administrator creates a cluster in the AWS console.

EC2 Configuration

The following table describes cloud provisioning configuration EC2 configuration properties:

Property	Description
EC2 Key Pair	EC2 key pair to enable communication with the EMR cluster master node. Optional. This credential enables you to log into the cluster. Configure this property if you intend the cluster to be non-ephemeral.
EC2 Subnet	ID of the subnet on the VPC in which to create the cluster. Use the subnet ID of the EC2 instance where the cluster runs.
Master Security Group	Optional. ID of the security group for the cluster master node. Acts as a virtual firewall to control inbound and outbound traffic to cluster nodes. Security groups are created when the AWS administrator creates and configures a cluster in a VPC. In the AWS console, the property is equivalent to ElasticMapReduce-master. You can use existing security groups, or the AWS administrator might create dedicated security groups for the ephemeral cluster. If you do not specify a value, the cluster applies the default security group for the VPC.
Additional Master Security Groups	Optional. IDs of additional security groups to attach to the cluster master node. Use a comma-separated list of security group IDs.
Core and Task Security Group	Optional. ID of the security group for the cluster core and task nodes. When the AWS administrator creates and configures a cluster In the AWS console, the property is equivalent to the ElasticMapReduce-slave security group If you do not specify a value, the cluster applies the default security group for the VPC.
Additional Core and Task Security Groups	Optional. IDs of additional security groups to attach to cluster core and task nodes. Use a comma-separated list of security group IDs.
Service Access Security Group	EMR managed security group for service access. Required when you provision an EMR cluster in a private subnet.

Azure Cloud Provisioning Configuration Properties

The properties in the Azure cloud provisioning configuration enable the Data Integration Service to contact and create resources on the Azure cloud platform.

Authentication Details

The following table describes authentication properties to configure:

Property	Description
Name	Name of the cloud provisioning configuration.
ID	ID of the cloud provisioning configuration. Default: Same as the cloud provisioning configuration name.
Description	Optional. Description of the cloud provisioning configuration.
Subscription ID	ID of the Azure account to use in the cluster creation process.

Property	Description
Tenant ID	A GUID string associated with the Azure Active Directory.
Client ID	A GUID string that is the same as the Application ID associated with the Service Principal. The Service Principal must be assigned to a role that has permission to create resources in the subscription that you identified in the Subscription ID property.
Client Secret	An octet string that provides a key associated with the client ID.

Storage Account Details

Choose to configure access to one of the following storage types:

- Azure Data Lake Storage (ADLS). See Azure documentation.
- An Azure Storage Account, known as general or blob storage. See Azure documentation.

The following table describes the information you need to configure Azure Data Lake Storage (ADLS) with the HDInsight cluster:

Property	Description
Azure Data Lake Store Name	Name of the ADLS storage to access. The ADLS storage and the cluster to create must reside in the same region.
Data Lake Service Principal Client ID	A credential that enables programmatic access to ADLS storage. Enables the Informatica domain to communicate with ADLS and run commands and mappings on the HDInsight cluster. The service principal is an Azure user that meets the following requirements: - Permissions to access required directories in ADLS storage. - Certificate-based authentication for ADLS storage. - Key-based authentication for ADLS storage.
Data Lake Service Principal Certificate Contents	The Base64 encoded text of the public certificate used with the service principal. Leave this property blank when you create the cloud provisioning configuration. After you save the cloud provisioning configuration, log in to the VM where the Informatica domain is installed and run infacmd ccps updateADLSCertificate to populate this property.
Data Lake Service Principal Certificate Password	Private key for the service principal. This private key must be associated with the service principal certificate.
Data Lake Service Principal Client Secret	An octet string that provides a key associated with the service principal.
Data Lake Service Principal OAUTH Token Endpoint	Endpoint for OAUTH token based authentication.

The following table describes the information you need to configure Azure General Storage, also known as blob storage, with the HDInsight cluster:

Property	Description
Azure Storage Account Name	Name of the storage account to access. Get the value from the Storage Accounts node in the Azure web console. The storage and the cluster to create must reside in the same region.
Azure Storage Account Key	A key to authenticate access to the storage account. To get the value from the Azure web console, select the storage account, then Access Keys. The console displays the account keys.

Cluster Deployment Details

The following table describes the cluster deployment properties that you configure:

Property	Description
Resource Group	Resource group in which to create the cluster. A resource group is a logical set of Azure resources.
Virtual Network Resource Group	Optional. Resource group to which the virtual network belongs. If you do not specify a resource group, the Data Integration Service assumes that the virtual network is a member of the same resource group as the cluster.
Virtual Network	Name of the virtual network or vnet where you want to create the cluster. Specify a vnet that resides in the resource group that you specified in the Virtual Network Resource Group property. The vnet must be in the same region as the region in which to create the cluster.
Subnet Name	Subnet in which to create the cluster. The subnet must be a part of the vnet that you designated in the previous property. Each vnet can have one or more subnets. The Azure administrator can choose an existing subnet or create one for the cluster.

External Hive Metastore Details

You can specify the properties to enable the cluster to connect to a Hive metastore database that is external to the cluster.

If you do not specify an existing external database in this dialog box, the cluster creates its own database on the cluster. This database is terminated when the cluster is terminated.

You can use an external relational database like MySQL or Amazon RDS as the Hive metastore database. The external database must be on the same cloud platform as the cluster to create.

The following table describes the Hive metastore database properties that you configure:

Property	Description
Database Name	Name of the Hive metastore database.
Database Server Name	Server on which the database resides. **Note:** The database server name on the Azure web console commonly includes the suffix `database.windows.net`. For example: `server123xyz.database.windows.net`. You can specify the database server name without the suffix and Informatica will automatically append the suffix. For example, you can specify `server123xyz`.

Property	Description
Database User Name	User name of the account for the domain to use to access the database.
Database Password	Password for the user account.

Databricks Cloud Provisioning Configuration Properties

The properties in the Databricks cloud provisioning configuration enable the Data Integration Service to contact and create resources on the Databricks cloud platform.

The following table describes the Databricks cloud provisioning configuration properties:

Property	Description
Name	Name of the cloud provisioning configuration. **Tip:** Because the Administrator tool lists cloud provisioning configuration objects with other connections, use a naming convention such as "CPC" as part of the name of the object to help identify it.
ID	The cluster ID of the Databricks cluster.
Description	Optional description of the cloud provisioning configuration.
Databricks domain	Domain name of the Databricks deployment.
Databricks token ID	The token ID created within Databricks required for authentication. **Note:** If the token has an expiration date, verify that you get a new token from the Databricks administrator before it expires.
Advanced Properties	Advanced properties that are unique to the Databricks cloud provisioning configuration.

Advanced Properties

Configure the following properties in the **Advanced Properties** of the Databricks configuration section:

infaspark.pythontx.exec

Required to run a Python transformation on the Databricks Spark engine. Set to the location of the Python executable binary on the worker nodes in the Databricks cluster.

When you provision the cluster at run time, set this property in the Databricks cloud provisioning configuration. Otherwise, set on the Databricks connection.

For example, set to:

```
infaspark.pythontx.exec=/databricks/python3/bin/python3
```

infaspark.pythontx.executorEnv.PYTHONHOME

Required to run a Python transformation on the Databricks Spark engine. Set to the location of the Python installation directory on the worker nodes in the Databricks cluster.

When you provision the cluster at run time, set this property in the Databricks cloud provisioning configuration. Otherwise, set on the Databricks connection.

For example, set to:

```
infaspark.pythontx.executorEnv.PYTHONHOME=/databricks/python3
```

Complete the Azure Cloud Provisioning Configuration

When you want to access Azure Data Lake Storage (ADLS) with the cluster workflow, complete the following steps after you configure and save the cloud provisioning configuration for Azure:

1. Log in to the VM where the Informatica domain is installed, and open a command shell.
2. From the command line, issue the following command:

```
/infacmd.sh ccps updateADLSCertificate -dn <domain name> -un <user name> -pd <password> -
cpcid <cloud provisioning connection name>
-certPath <domain location of certificate>
```

The command automatically populates the Data Lake Service Principal Certificate Contents property of the cloud provisioning connection.

Create a Hadoop Cluster Configuration

Import cluster information into the domain. When you import cluster information, you import values from *-site.xml files to create a domain object called a cluster configuration.

When you create a cluster configuration, choose one of the following options to import cluster properties:

Import from cluster

When you import directly from the cluster, you enter cluster connection information. The Service Manager uses the information to connect to the cluster and get cluster configuration properties.

Note: You can import directly from Azure HDInsight. Amazon EMR does not support direct import.

Import from file

When you import from a file, you browse to an archive file that the cluster administrator created. Use this option if the cluster administrator requires you to do so.

Note: If you import from an Amazon EMR cluster, you must import from a file.

Before You Import

Before you can import the cluster configuration, you must get information from the Hadoop administrator, based on the method of import.

If you import directly from the cluster, contact the Hadoop administrator to get cluster connection information. If you import from a file, get an archive file of exported cluster information.

For more information about required cluster information, see the *Data Engineering Integration Guide*

Importing a Hadoop Cluster Configuration from the Cluster

When you import the Hadoop cluster configuration directly from the cluster, you provide information to connect to the cluster.

Get cluster connection information from the Hadoop administrator.

1. From the **Connections** tab, click the **ClusterConfigurations** node in the Domain Navigator.
2. From the Actions menu, select **New** > **Cluster Configuration**.

The **Cluster Configuration** wizard opens.

3. Configure the following General properties:

Property	Description
Cluster configuration name	Name of the cluster configuration.
Description	Optional description of the cluster configuration.
Distribution type	The cluster Hadoop distribution type.
Method to import the cluster configuration	Choose **Import from cluster**.
Create connections	Choose to create Hadoop, HDFS, Hive, and HBase connections. If you choose to create connections, the Cluster Configuration wizard associates the cluster configuration with each connection that it creates. The Hadoop connection contains default values for properties such as cluster environment variables, cluster path variables, and advanced properties. Based on the cluster environment and the functionality that you use, you can add to the default values or change the default values of these properties. For a list of Hadoop connection properties to configure, see Hadoop Cluster Connection Properties. If you do not choose to create connections, you must manually create them and associate the cluster configuration with them. **Important:** When the wizard creates the Hive connection, it populates the Metadata Connection String and the Data Access Connection String properties with the value from the hive.metastore.uris property. If the Hive metastore and HiveServer2 are running on different nodes, you must update the Metadata Connection String to point to the HiveServer2 host.

The cluster properties appear.

4. Configure the following properties:

Property	Description
Host	IP address of the cluster manager.
Port	Port of the cluster manager.
User ID	Cluster user ID.
Password	Password for the user.
Cluster name	Name of the cluster. Use the display name if the cluster manager manages multiple clusters. If you do not provide a cluster name, the wizard imports information based on the default cluster.

5. Click **Next** and verify the cluster configuration information on the summary page.

Importing a Hadoop Cluster Configuration from a File

You can import properties from an archive file to create a cluster configuration.

Before you import from the cluster, you must get the archive file from the Hadoop administrator.

1. From the **Connections** tab, click the **ClusterConfigurations** node in the Domain Navigator.

2. From the Actions menu, select **New** > **Cluster Configuration**.

   The **Cluster Configuration** wizard opens.

3. Configure the following properties:

Property	Description
Cluster configuration name	Name of the cluster configuration.
Description	Optional description of the cluster configuration.
Distribution type	The cluster Hadoop distribution type.
Method to import the cluster configuration	Choose **Import from file** to import properties from an archive file.
Create connections	Choose to create Hadoop, HDFS, Hive, and HBase connections. If you choose to create connections, the Cluster Configuration wizard associates the cluster configuration with each connection that it creates. The Hadoop connection contains default values for properties such as cluster environment variables, cluster path variables, and advanced properties. Based on the cluster environment and the functionality that you use, you can add to the default values or change the default values of these properties. For a list of Hadoop connection properties to configure, see Hadoop Cluster Connection Properties. If you do not choose to create connections, you must manually create them and associate the cluster configuration with them. **Important:** When the wizard creates the Hive connection, it populates the Metadata Connection String and the Data Access Connection String properties with the value from the hive.metastore.uris property. If the Hive metastore and HiveServer2 are running on different nodes, you must update the Metadata Connection String to point to the HiveServer2 host.

4. Click **Browse** to select a file. Select the file and click **Open**.
5. Click **Next** and verify the cluster configuration information on the summary page.

Create the Cluster Configuration

Perform the following steps to create a cluster configuration:

1. In the Administrator tool, open the **Connections** tab.
2. In the Domain navigator, select **Cluster Configurations**.
3. Click the **Actions** dropdown control and select **New** > **Cluster Configuration**.
4. Enter property values in the configuration wizard, and then click **Finish**.

Note: If you want to use ADLS storage with an Azure HDInsight cluster, you must run infacmd ccps updateADLSCertificate to populate the Data Lake Service Principal Certificate Contents property after you create the Azure cloud provisioning configuration.

Create a Databricks Cluster Configuration

A Databricks cluster configuration is an object in the domain that contains configuration information about the Databricks cluster. The cluster configuration enables the Data Integration Service to push mapping logic to the Databricks environment.

Use the Administrator tool to import configuration properties from the Databricks cluster to create a cluster configuration. You can import configuration properties from the cluster or from a file that contains cluster properties. You can choose to create a Databricks connection when you perform the import.

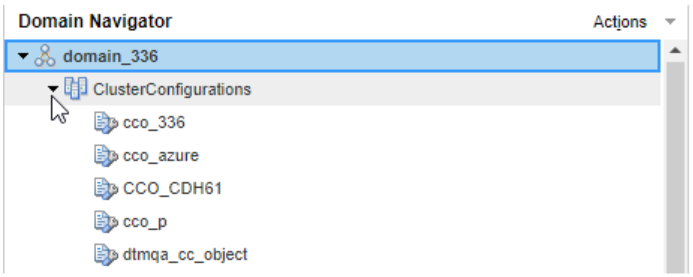
Note: Ensure that you integrate a Databricks cluster with only one Informatica domain.

Importing a Databricks Cluster Configuration from the Cluster

When you import the cluster configuration directly from the cluster, you provide information to connect to the cluster.

Before you import the cluster configuration, get cluster information from the Databricks administrator.

1. From the **Connections** tab, click the **ClusterConfigurations** node in the Domain Navigator.
   The following image shows the **ClusterConfigurations** node in the Domain Navigator on the **Connections** tab:

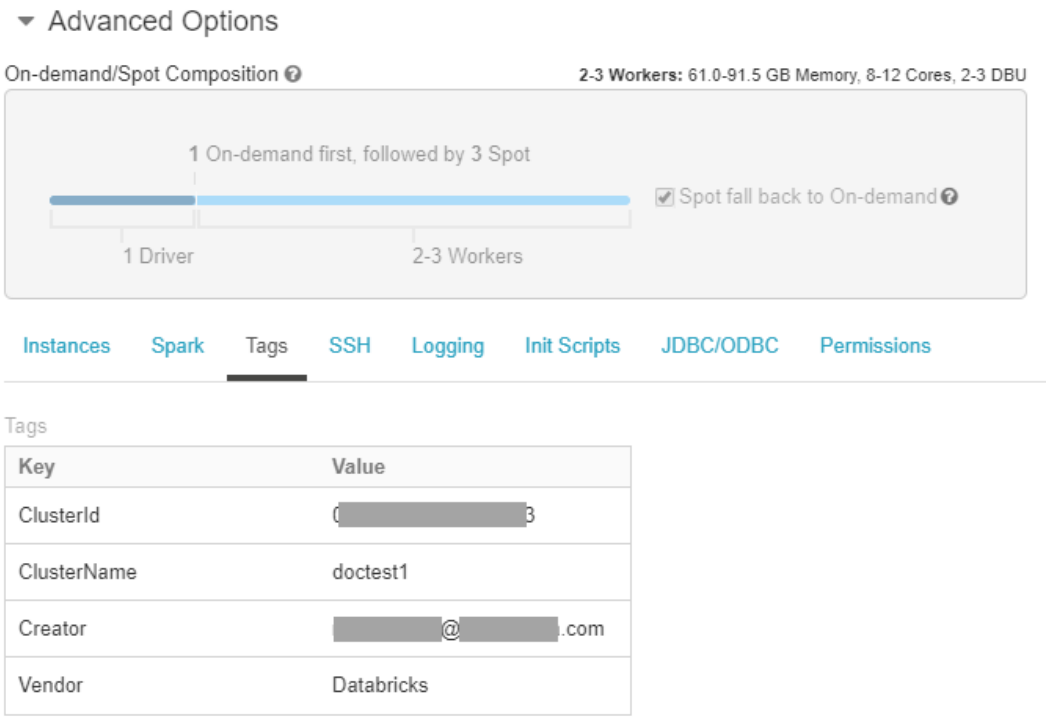


2. From the Actions menu, select **New** > **Cluster Configuration**.
   The **Cluster Configuration** wizard opens.
3. Configure the following properties:

Property	Description
Cluster configuration name	Name of the cluster configuration.
Description	Optional description of the cluster configuration.
Distribution type	The distribution type. Choose **Databricks**.
Method to import the cluster configuration	Choose **Import from cluster**.
Databricks domain	Domain name of the Databricks deployment.
Databricks access token	The token ID created within Databricks, required for authentication. . The token ID is visible only when it is created by the Databricks administrator. See Prepare a Databricks User for Token Identification. **Note:** If the token has an expiration date, verify that you get a new token from the Databricks administrator before it expires.

Property	Description
Databricks cluster ID	The cluster ID of the Databricks cluster. To find the cluster ID on the Databricks portal, follow these steps: 1. Select **Clusters** from the object bar on the left side. 2. Select the cluster you want to integrate with the Informatica domain. 3. Click the **Spark ID** tab and expand the list of **Spark Properties**. 4. Select the **Tags** tab. See the image below.
Create connection	Choose to create a Databricks connection. If you choose to create a connection, the **Cluster Configuration** wizard associates the cluster configuration with the Databricks connection. If you do not choose to create a connection, you must manually create one and associate the cluster configuration with it. To manually create a connection, see the section about manually creating a connection in the Data Integration Engineering Administrator Guide.

To find the ClusterID, expand Spark **Advanced Properties** and then select the **Tags** tab. The following image shows the redacted ClusterID on the **Tags** tab:



4. Click **Next** to verify the information on the summary page.

Importing a Databricks Cluster Configuration from a File

You can import properties from an archive file to create a cluster configuration.

Complete the following tasks to import a Databricks cluster from a file:

1. Get required cluster properties from the Databricks administrator.
2. Create an .xml file with the cluster properties, and compress it into a .zip or .tar file.

3. Log in to the Administrator tool and import the file.

Create the Import File

To import the cluster configuration from a file, you must create an archive file.

To create the .xml file for import, you must get required information from the Databricks administrator. You can provide any name for the file and store it locally.

The following table describes the properties required to import the cluster information:

Property Name	Description
cluster_name	Name of the Databricks cluster.
cluster_ID	The cluster ID of the Databricks cluster.
baseURL	URL to access the Databricks cluster. This is the URL that appears in your browser menu bar when you access the Databricks cluster.
accesstoken	The token ID created within Databricks required for authentication.

Optionally, you can include other properties specific to the Databricks environment.

The following text shows a sample import file with the required properties:

```
<?xml version="1.0" encoding="UTF-8"?><configuration>
  <property>
    <name>cluster_name</name>
    <value>my_cluster</value>
  </property>
  <property>
    <name>cluster_id</name>
    <value>0926-294544-bckt123</value>
  </property>
  <property>
    <name>baseURL</name>
    <value>https://<AWS Databricks domain></value>
  </property>
  <property>
    <name>accesstoken</name>
    <value>dapicf76c2d4567c6sldn654fe875936e778</value>
  </property>
</configuration>
```

When you complete the .xml file, compress it into a .zip or .tar file for import.

Import the Cluster Configuration

After you create the .xml file with the cluster properties, use the Administrator tool to import into the domain and create the cluster configuration.

1. From the **Connections** tab, click the **ClusterConfigurations** node in the Domain Navigator.
2. From the Actions menu, select **New** > **Cluster Configuration**.
   The **Cluster Configuration** wizard opens.

3. Configure the following properties:

Property	Description
Cluster configuration name	Name of the cluster configuration.
Description	Optional description of the cluster configuration.
Distribution type	The distribution type. Choose **Databricks**.
Method to import the cluster configuration	Choose **Import from file**.
Upload configuration archive file	The full path and file name of the file. Click the Browse button to navigate to the file.
Create connection	Choose to create a Databricks connection. If you choose to create a connection, the **Cluster Configuration** wizard associates the cluster configuration with the Databricks connection. If you do not choose to create a connection, you must manually create one and associate the cluster configuration with it.

4. Click **Next** to verify the information on the summary page.

Databricks Connection Properties

Use the Databricks connection to run mappings on a Databricks cluster.

A Databricks connection is a cluster type connection. You can create and manage a Databricks connection in the Administrator tool or the Developer tool. You can use infacmd to create a Databricks connection. Configure properties in the Databricks connection to enable communication between the Data Integration Service and the Databricks cluster.

The following table describes the general connection properties for the Databricks connection:

Property	Description
Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Optional. The description of the connection. The description cannot exceed 4,000 characters.
Connection Type	Choose Databricks.
Cluster Configuration	Name of the cluster configuration associated with the Databricks environment. Required if you do not configure the cloud provisioning configuration.

Property	Description
Cloud Provisioning Configuration	Name of the cloud provisioning configuration associated with a Databricks cloud platform. Required if you do not configure the cluster configuration.
Staging Directory	The directory where the Databricks Spark engine stages run-time files. If you specify a directory that does not exist, the Data Integration Service creates it at run time. If you do not provide a directory path, the run-time staging files are written to */<cluster staging directory>/DATABRICKS*.
Advanced Properties	List of advanced properties that are unique to the Databricks environment. You can configure run-time properties for the Databricks environment in the Data Integration Service and in the Databricks connection. You can override a property configured at a high level by setting the value at a lower level. For example, if you configure a property in the Data Integration Service custom properties, you can override it in the Databricks connection. The Data Integration Service processes property overrides based on the following priorities: 1. Databricks connection advanced properties 2. Data Integration Service custom properties **Note:** Informatica does not recommend changing these property values before you consult with third-party documentation, Informatica documentation, or Informatica Global Customer Support. If you change a value without knowledge of the property, you might experience performance degradation or other unexpected results.

Advanced Properties

Configure the following properties in the **Advanced Properties** of the Databricks configuration section:

infaspark.json.parser.mode

Specifies the parser how to handle corrupt JSON records. You can set the value to one of the following modes:

- DROPMALFORMED. The parser ignores all corrupted records. Default mode.
- PERMISSIVE. The parser accepts non-standard fields as nulls in corrupted records.
- FAILFAST. The parser generates an exception when it encounters a corrupted record and the Spark application goes down.

infaspark.json.parser.multiLine

Specifies whether the parser can read a multiline record in a JSON file. You can set the value to true or false. Default is false. Applies only to non-native distributions that use Spark version 2.2.x and above.

infaspark.flatfile.writer.nullValue

When the Databricks Spark engine writes to a target, it converts null values to empty strings (" "). For example, 12, AB,"",23p09udj.

The Databricks Spark engine can write the empty strings to string columns, but when it tries to write an empty string to a non-string column, the mapping fails with a type mismatch.

To allow the Databricks Spark engine to convert the empty strings back to null values and write to the target, configure the property in the Databricks Spark connection.

Set to: TRUE

infaspark.pythontx.exec

Required to run a Python transformation on the Databricks Spark engine. Set to the location of the Python executable binary on the worker nodes in the Databricks cluster.

When you provision the cluster at run time, set this property in the Databricks cloud provisioning configuration. Otherwise, set on the Databricks connection.

For example, set to:

```
infaspark.pythontx.exec=/databricks/python3/bin/python3
```

infaspark.pythontx.executorEnv.PYTHONHOME

Required to run a Python transformation on the Databricks Spark engine. Set to the location of the Python installation directory on the worker nodes in the Databricks cluster.

When you provision the cluster at run time, set this property in the Databricks cloud provisioning configuration. Otherwise, set on the Databricks connection.

For example, set to:

```
infaspark.pythontx.executorEnv.PYTHONHOME=/databricks/python3
```

Create a Cluster Connection

Create a dedicated cluster connection, such as Hadoop or Databricks, to use with the cluster workflow.

The cluster connection saves property values for the Data Integration Service to use for cluster workflow operations. When you run a cluster workflow, the Data Integration Service uses settings in the cluster connection to run jobs on the cluster.

When you configure the connection, populate the Cloud Provisioning Configuration property with the name of the cloud provisioning configuration that you created for the workflow. Leave the Cluster Configuration property blank.

When you create the workflow, populate the Connection Name property of the Create Cluster task with this cluster connection.

For information about creating connections in the Administrator tool, see the Connections Reference in the *Data Engineering Administrator Guide*.

Developer Tasks

After you complete administrator tasks, complete the following tasks in the Developer tool:

Step 1. Create the cluster workflow.

Create the cluster workflow and add and configure workflow elements.

Step 2. Add Mapping and other tasks to the workflow.

Populate the cluster workflow with at least one Mapping task. You can add Command or other workflow tasks and events. Set mapping properties and Mapping task properties to specify where the workflow runs Mapping tasks.

Step 3. Optional: Add a Delete Cluster task.

Add a Delete Cluster task to the workflow to save cloud platform resources when workflow tasks are complete. The Delete Cluster task terminates and deletes the cluster and the Hadoop connections that connect to it. If a cluster workflow does not include a Delete Cluster task, the cluster remains when all workflow tasks are complete.

A cluster created by a workflow and terminated with a Delete Cluster task is called an ephemeral cluster.

Step 4. Deploy and run the workflow.

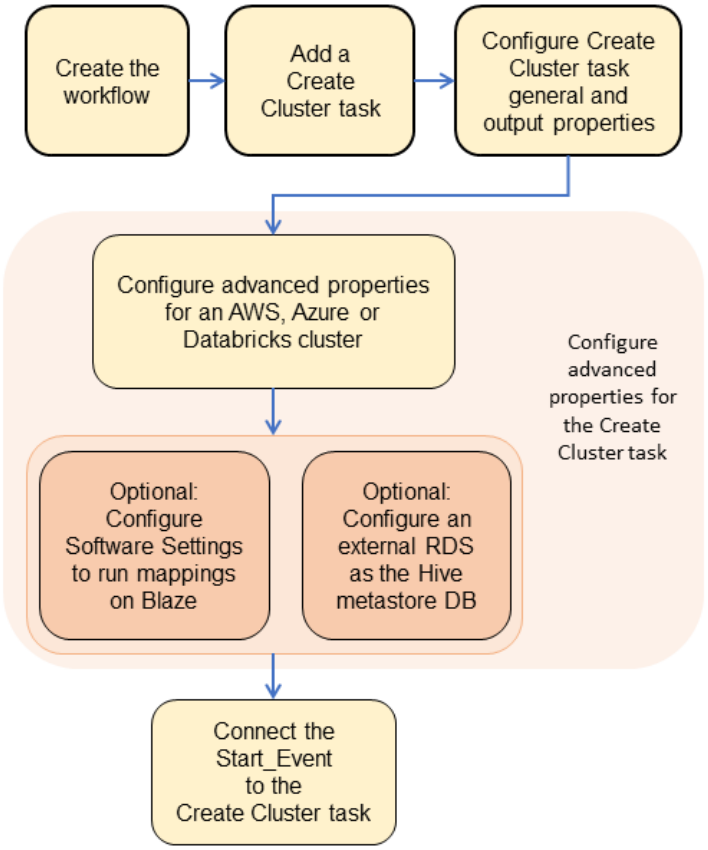
After you complete development of the cluster workflow, deploy and run the workflow.

Step 1. Create the Cluster Workflow and Configure the Create Cluster Task

Create the cluster workflow and add and configure workflow elements.

The cluster workflow must have a Create Cluster task and at least one Mapping task. You can add Command and other workflow tasks. You can also add a Delete Cluster task, so that the cluster becomes an ephemeral cluster.

The following image shows the steps to create the workflow and to create and configure the Create Cluster task:



Configure the Cluster Workflow

A cluster workflow must have one Create Cluster task.

1. In the Developer tool, create a workflow.
2. From the palette of tasks, drag a Create Cluster task to the workflow editor.
3. Complete the Create Cluster task general properties.

Property	Description
Name	Task name.
Description	Optional description.

Property	Description
Connection Name	Name of the cloud provisioning configuration to use with the workflow.
Connection Type	Choose from the following options: - Amazon EMR. Create an Amazon EMR cluster. - HDInsight. Create an Azure HDInsight cluster.

4. Configure task input and output properties.

   Input properties

   The Create Cluster task does not require any unique values for task input properties.

   Output properties

   Set the Cluster Identifier property to the default value, AutoDeployCluster.

   Note: The Cluster Identifier property of the Create Cluster task overrides the Cluster Identifier property of the Mapping task.

5. Set the advanced properties that correspond to your cloud platform.
   - To create a cluster on Amazon Web Services, see "Advanced Properties for Amazon EMR" on page 27.
   - To create a cluster on Microsoft Azure, see "Advanced Properties for Azure HDInsight" on page 30.
6. Configure the Software Settings property in the advanced properties if you want to perform the following optional tasks:
   - To run mappings on the Blaze engine, see "Advanced Properties for the Blaze Engine" on page 36.
   - To configure an external relational database as the Hive metastore database, see "Advanced Properties for a Hive Metastore Database" on page 37.
7. Connect the workflow Start_Event to the Create Cluster task.

Step 2. Create Mapping and Other Workflow Tasks

Populate the cluster workflow with at least one Mapping task. You can add Command or other workflow tasks and events.

Before you create a Mapping task, configure mappings to run in the cluster workflow. Then add Mapping tasks to run the mappings that you prepared.

You can also add Command and other tasks to the workflow.

Prepare Mappings for the Cluster Workflow

Prepare mappings for inclusion in the cluster workflow by setting mapping and Mapping task properties to specify where mappings run.

You can create a cluster workflow to run some mappings on the cluster that the workflow creates, and other mappings on another cluster.

Run the mapping on the cluster that the workflow creates.

The following table describes the run-time behavior based on how you configure the mapping and Mapping task:

Mapping Hadoop Connection Property	Mapping Task Cluster Identifier Property	Run Time Behavior
Auto Deploy	Auto Deploy	The Data Integration Service generates temporary Hadoop connections based on the values in the Hadoop connection associated with the workflow, and uses the temporary connections to run mappings on the cluster.
<Hadoop connection name>	Auto Deploy	The Mapping task Cluster Identifier property overrides the mapping Hadoop connection property. You might want to do this if you want to maintain the mapping Hadoop connection property value.

Run the mapping on another cluster.

The following table describes the run-time behavior based on how you configure the mapping and Mapping task:

Mapping Hadoop Connection Property	Mapping Task Cluster Identifier Property	Run Time Behavior
<Hadoop connection name>	Blank	The Mapping task Cluster Identifier property takes input from the Hadoop connection and runs the mapping on the cluster identified in the cloud configuration property of the Hadoop connection.

Add Mapping and Other Tasks to the Workflow

1. Drag a Mapping task from the task list to the workflow editor.

   The **Mapping Task** dialog box opens.
2. Name the Mapping task.
3. Select a mapping to run with the Mapping task. Click **Browse** next to the Mapping property, select a mapping, and click **Finish**.
4. Optionally select a parameter set to associate with the Mapping task. Click **Browse** next to the Parameter Set property, select a parameter set, and click **Finish**.

   For more information on how to use parameter sets with mappings, see the *Informatica Developer Mapping Guide*.
5. Optionally complete Input and Output properties.

   The Mapping task does not require any unique values for input or output properties.
6. Configure the Cluster Identifier property in Advanced properties.

   The Cluster Identifier property designates the cluster to use to run the Mapping task.

The following table describes values for Cluster Identifier properties:

Value	Description
Blank (no value)	Run the mapping runs on the cluster configured in the Hadoop connection associated with the mapping.
AutoDeploy	Run the mapping on the cluster that the workflow creates. When you choose this option, it also populates the Cluster Identifier property in the Create Cluster task with the value Set to AutoDeployCluster. Default is AutoDeploy.
(Assign to task input)	Select this option to accept input from another source than the Create Cluster task. If you choose this option, enter a parameter value in the Cluster Identifier property of the Mapping task Input properties tab.

7. Click **Finish** to the Mapping task.
8. Optionally add more Mapping and other tasks to the workflow.

   You can include any other workflow tasks in a cluster workflow. For example, you might want to add a Command task to perform tasks after a Mapping task runs.

Mapping Task Properties

Set mapping and Mapping task properties to specify where the workflow runs Mapping tasks.

When you include a Mapping task in a cluster workflow, you must identify the cluster where you want the mapping to run.

The following table describes the Cluster Identifier property values that you can configure in the Advanced properties of a Mapping task:

Cluster Identifier Property	Description
Auto Deploy	Runs the mapping on the cluster that the workflow creates. The Data Integration Service generates a temporary cluster connection based on the connection associate with the cluster provisioning configuration. The connection property in the mapping must also be set to Auto Deploy.
Blank	Runs the mapping on the cluster configured in the cluster connection associated with the mapping. The connection property in the mapping must be set to a cluster connection associated with a cluster configuration. Validation fails if it is associated with the cluster provisioning configuration.
Assign to task input	Accept input from a source other than the Create Cluster task. Use this option if you configure input properties to use parameters.

Note: If the Mapping task and the Create Cluster task contain different values for the cluster identifier, the Create Cluster task value takes precedence.

Step 3. Add a Delete Cluster Task

To create an ephemeral cluster, add a Delete Cluster task.

The Delete Cluster task terminates the cluster and deletes the cluster and other resources that the cluster workflow creates.

If you do not add a Delete Cluster task, the cluster that the workflow creates remains running when the workflow ends. You can delete the cluster at any time.

1. Drag a Delete Cluster task to the workflow editor.
2. In the General properties, optionally rename the Delete Cluster task.
3. Connect the final task in the workflow to the Delete Cluster task, and connect the Delete Cluster task to the workflow End_Event.

You can also use infacmd ccps deleteCluster to delete a cloud cluster.

Step 4. Deploy and Run the Workflow

After you complete the cluster workflow, deploy and run the workflow.

You can monitor cluster provisioning tasks on the AWS or Azure web console. If you configured a log location, view the logs at the location that you configured in the Create Cluster task properties.

You can also monitor Data Integration Service jobs in the Administrator tool.

Note: After the workflow begins executing tasks, the task to provision the cluster may take several minutes.

Monitoring Azure HDInsight Cluster Workflow Jobs

You can access mapping log URLs through the **Monitoring** tab in the Administrator tool to monitor workflow jobs that run on an Azure HDInsight cluster. The log location depends on the run-time engine that each mapping uses.

To access the monitoring URL for mappings that run on the Blaze or Spark engine, expand the workflow and the mapping on the **Monitoring** tab. Select the Grid Task and view the value for the Monitoring URL property in the lower pane. Use this path to find the log.

Create Cluster Task Properties

When you configure a Create Cluster task, you configure options that are common for all cluster types, and you configure advanced options for specific clusters.

General Properties for the Create Cluster Task

The following table describes the General properties for the Create Cluster task:

Property	Description
Name	Task name.
Description	Optional description.

Property	Description
Connection Name	Name of the cloud provisioning configuration to use with the workflow.
Connection Type	Choose from the following options: - Amazon EMR. Create an Amazon EMR cluster in the Hadoop environment. - HDInsight. Create an Azure HDInsight cluster in the Hadoop environment. - Databricks. Create a Databricks cluster in the Databricks environment.

Input Properties for the Create Cluster Task

You can assign task input to workflow parameters and variables.

The following table describes the Input properties for the Create Cluster task:

Property	Description
Create Cluster Task Inputs	Parameters or variables to use for task input.

Output Properties for the Create Cluster Task

You can assign variables to the Create Cluster task output.

The following table describes the Output properties for the Create Cluster task:

Property	Description
Start Time	Date and time that the task started running.
End Time	Date and time that the task stopped running
Is Successful	Indicates whether the task ran successfully.
Cluster Identifier	The Cluster Identifier property of the Create Cluster task overrides the Cluster Identifier property of the Mapping task. Default is AutoDeployCluster.

Advanced Properties for the Create Cluster Task

Advanced properties for the Create Cluster task depend on the cluster type that you create.

For information about advanced properties for Microsoft Azure, see "Advanced Properties for Azure HDInsight" on page 30.

For information about advanced properties for Amazon Web Services, see "Advanced Properties for Amazon EMR" on page 27.

For information about advanced properties to run mappings on the Blaze engine, see "Advanced Properties for the Blaze Engine" on page 36.

For information about advanced properties for Databricks, see "Advanced Properties for Azure Databricks" on page 34.

Advanced Properties for Amazon EMR

Set the advanced properties for an Amazon EMR cluster.

General Options

The following table describes general options that you can set for an EMR cluster:

Property	Description
Cluster Name	Name of the cluster to create.
Release Version	EMR version to run on the cluster. Enter the AWS version tag string to designate the version. For example: `emr-5.8.0` Default is Latest version supported.
Connection Name	Name of the Hadoop connection that you configured for use with the cluster workflow.
S3 Log URI	Optional. S3 location of logs for cluster creation. Format: `s3://<bucket name>/<folder name>` If you do not supply a location, no cluster logs will be stored.

Master Instance Group Options

The following table describes master instance group options that you can set for an EMR cluster:

Property	Description
Master Instance Type	Master node EC2 instance type. You can specify any available EC2 instance type. Default is m4.4xlarge.
Master Instance Maximum Spot Price	Maximum spot price for the master node. Setting this property changes the purchasing option of the master instance group to Spot instead of On-demand.

Core Instance Group Options

The following table describes core instance group options that you can set for an EMR cluster:

Property	Description
Core Instance Type	Core node EC2 instance type. You can specify any available EC2 instance type. Default is m4.4xlarge.
Core Instance Count	Number of core EC2 instances to create in the cluster. Default is 2.
Core Instance Maximum Spot Price	Maximum spot price for core nodes. Setting this property changes the purchasing option of the core instance group to Spot instead of On-demand.
Core Auto-Scaling Policy	Optional. Auto-scaling policy for core instances. Type the policy JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_policy_config_file>`

Task Instance Group Options

The following table describes task instance group options that you can set for an EMR cluster:

Property	Description
Task Instance Type	Task node EC2 instance type. You can specify any available EC2 instance type. Default is m4.4xlarge.
Task Instance Count	Number of task EC2 instances to create in the cluster. Default is 2.
Task Instance Maximum Spot Price	Maximum spot price for task nodes. Setting this property changes the purchasing option of the task instance group to Spot instead of On-demand.
Task Auto-Scaling Policy	Optional. Auto-scaling policy for task instances. Type the policy JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_policy_config_file>`

Additional Options

The following table describes additional options that you can set for an EMR cluster:

Property	Description
Root device EBS volume size (GB)	Number of GB of the EBS root device volume. Enter a value between 10 and 100. Default is 10.
Tags	Optional. Tags to propagate to cluster EC2 instances. Tags assist in identifying EC2 instances. Format: `TagName1=TagValue1,TagName2=TagValue2`
Bootstrap Actions	Optional. Actions to perform after EC2 instances are running, and before applications are installed. Type the JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_policy_config_file>`
Custom AMI ID	Optional. ID of a custom Amazon Linux Amazon Machine Image (AMI). Copy the value from the AWS console.
Security Configuration	Optional. The name of a security configuration for authentication and encryption on the cluster. Amazon EMR supports server-side encryption (SSE) and client-side encryption (CSE) configurations. You can use the following at-rest security configurations: - SSE with Amazon S3-managed keys (SSE-S3) - SSE with AWS KMS-managed keys (SSE-KMS) - CSE with AWS KMS-managed keys (CSE-KMS) - Custom CSE configurations* You can use the following in-transit security configurations: - PEM - Custom in-transit configurations* You can also use custom AMIs for local disk security. ** To use custom security configurations, manually copy the .jar file to the Data Integration Service machine.*

Property	Description
Applications	Optional. Applications to add to the default applications that AWS installs. AWS installs certain applications when it creates an EMR cluster. In addition, you can specify additional applications. Select additional applications from the drop-down list. This field is equivalent to the Software Configuration list in the AWS EMR cluster creation wizard.
Software Settings	Optional. Custom configurations to apply to the applications installed on the cluster. This field is equivalent to the Edit Software Settings field in the AWS cluster creation wizard. You can use this as a method to modify the software configuration on the cluster. Type the configuration JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_custom_config_file>`
Steps	Optional. Commands to run after cluster creation. For example, you can use this to run Linux commands or HDFS or Hive Hadoop commands. This field is equivalent to the Add Steps field in the AWS cluster creation wizard. Type the command statement here, or or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_command_file>`

Advanced Properties for Azure HDInsight

Set the advanced properties for an Azure HDInsight cluster.

The following table describes the Advanced properties for a Microsoft Azure HDInsight cluster:

Property	Description
Cluster Name	Name of the cluster to create.
Azure Cluster Type	Type of the cluster to be created. Choose one of the options in the drop-down list. Default is Hadoop.
HDInsight version	HDInsight version to run on the cluster. Enter the HDInsight version tag string to designate the version. Default is the latest version supported.
Azure Cluster Location	Use the drop-down list to choose the location in which to create the cluster.
Head Node VM Size	Size of the head node instance to create. Default is Standard_D12_v2.
Number of Worker Node Instances	Number of worker node instances to create in the cluster. Default is 2.
Worker Node VM Size	Size of the worker node instance to create. Default is Standard_D13_v2.

Property	Description
Default Storage Type	Primary storage type to be used for the cluster. Choose one of the following options: - Azure Data Lake Store - Azure BLOB storage account Default is BLOB storage
Default Storage Container or Root Mount Path	Default container for data. Type one of the following paths: - For ADLS storage, type the path to the storage. For example, you can type `storage-name` or `storage-name/folder-name`. - For blob storage, type the path to the container. Format: `/path/`
Log Location	Optional. Path to the directory to store workflow event logs. Default is `/app-logs`.
Attach External Hive Metastore	If you select this option, the workflow attaches an external Hive metastore to the cluster if you configured an external Hive metastore in the cloud provisioning configuration.

Bootstrap JSON String	JSON statement to run during cluster creation. You can use this statement to configure cluster details. For example, you could configure Hadoop properties on the cluster, add tags to cluster resources, or run script actions.
	Choose one of the following methods to populate the property:
	- Type the JSON statement. Use the following format:
	`{ "core-site" : { "<sample_property_key1>": "<sample_property_val1>", "<sample_property_key2>": "<sample_property_val2>" }, "tags": { "<tag_key>": "<tag_val>" }, "scriptActions": [ { "name": "setenvironmentvariable", "uri": "scriptActionUri", "parameters": "headnode" } ] }`
	- Provide a path to a file that contains a JSON statement. Format:
	`file://<path_to_bootstrap_file>`

Using ADLS Gen2 Resources with a Cluster Workflow

When a cluster workflow accesses ADLS Gen2 resources, verify that the Data Integration Service machine is configured to run in the same time zone that the cluster runs in.

Set the time zone environment variable on each Data Integration Service machine to the time zone of the cluster and then restart the domain.

To set the time zone, run the following command on the Data Integration Service machine:

```
setenv TZ <cluster time zone>
```

Advanced Properties for AWS Databricks

Set the advanced properties for an AWS Databricks cluster.

General Options

The following table describes the general options that you can set for a Databricks cluster:

Property	Description
Cluster Name	Name of the cluster to create.
Databricks Runtime Version	The Databricks version to run on the cluster. Default is the latest supported version. To manually enter a version, select *(Assign to Task Input)*. The Developer tool creates a Create Cluster Task Input called *Databricks Runtime Version*. Set the value of the input to the Databricks version tag string. For example, set the value to `5.5.x-scala2.11` to use Databricks version 5.5.
Python Version	The Python version to run if you include a Python transformation. Default is the latest supported version.
Warm Pool	Check this option to enable creation of the Databricks cluster using a warm pool. Default is unchecked (false).
Pool ID	Value of the DatabricksInstancePoolId property that identifies the warm pool resource. To retrieve the value of this property, go to the warm pool in the Databricks workspace and browse to **Configuration** > **Tags**. Required when the Warm Pool option is checked.
Driver Type	The type of node that you want to use for the driver node. Default is the worker type ID.
Worker Type	The type of node that you want to use for the worker node.
Workers	The number of worker nodes to create for the cluster. If you configure the cluster to scale automatically, this property is ignored. Default is 1.
Autoscale	Automatically scales the number of worker nodes based on workload.
Min Workers	The minimum number of worker nodes to use when the cluster is configured to scale automatically. Default is 0.
Max Workers	The maximum number of worker nodes to use when the cluster is configured to scale automatically. Default is 1.

Advanced Options

Configure advanced options such as automatic termination and on-demand instances.

The following table describes advanced options that you can set for an AWS Databricks cluster:

Property	Description
Enable autoscaling local storage	Enables Databricks to monitor available disk space on worker nodes and automatically add additional EBS volumes.
EBS Volume Type	The type of volume that Databricks can add to cluster nodes. Set this property when you enable autoscaling.
Number of Volumes	The number of volumes to provision for each instance. Enter a value between 0 and 10. Set this property when you enable autoscaling and configure EBS volume type.
Size in GB	The size in gigabytes of each EBS volume. Set this property when you enable autoscaling and configure EBS volume type.
Auto Termination	Enables automatic termination of the cluster.
Auto Terminate Time	Terminates the cluster after it is inactive for the specified number of minutes. Enter a value between 10 and 10,000. If you do not configure this, or if you set to 0, the cluster will not automatically terminate.
On-Demand/Spot Composition	The number of on-demand nodes. Enter a value between 0 and the number of worker nodes set in General Options. Any remaining worker nodes will be spot instances. On-demand nodes are always available to use. Spot instances might terminate running jobs if they become unavailable. The driver node is always an on-demand node. Set this property when you enable **Spot fall back to On-Demand**. Default is 1.
Spot fall back to On-Demand	Enables on-demand instances to be used as a fallback. If you are using spot instances and the market price for spot instances surges above your spot bid price, AWS terminates the spot instances. When you enable this property, on-demand instances are used in place of the spot instances when they terminate.
Availability Zone	The AWS cluster availability zone. Default is us-east-1e.
Spot Bid Price	The maximum percent of the on-demand instance price that you bid on spot instances. Spot instances are priced as a percentage of the on-demand price and are not always available. If the market price for spot instances surges above the bid price set here and you do not enable **Spot fall back to On-Demand**, AWS terminates the spot instance. Default is 100%.
IAM Role ARN	The instance profile ARN (Amazon Resource Name) that corresponds to the AWS IAM (Identity and Access Management) role. Copy the value from the AWS console in the following format: `arn:aws:iam::<account-id>:instance-profile/<role-name>` IAM roles allow you to access data from Databricks clusters. Add new IAM roles in the Administrator tool.

Property	Description
Spark Configurations	Performance configurations for the Databricks Spark engine. Enter key-value pairs in the following format: key1='value1' key2='value2'. You can also provide a path to a file that contains the key-value pairs.
Environment Variables	Environment variables that you can configure for the Databricks Spark engine. Enter key-value pairs in the following format: key1='value1' key2='value2'
Cluster Tags	Labels that you can assign to resources for tracking purposes. Enter key-value pairs in the following format: <key1>=<value1>,<key2>=<value2>. You can also provide a path to a local file that contains the key-value pairs. Use the following format: `file:\\<file path>`
SSH Public Key	The SSH public key to log into the driver and worker instances if you enable SSH. Copy the value from the Databricks console.
Cluster Log Conf	The location to deliver logs for long-term storage. If configured, the Databricks Spark engine will deliver the logs every five minutes. Provide the path to DBFS.
Init Scripts	The location where you store init scripts. You can enter multiple destinations. The scripts are run sequentially in the order that you configure them. If you need to install additional Python libraries, specify the init script file location in this property. Use the following format: `dbfs:/<path to init script>,dbfs:/<path to init script>`

Advanced Properties for Azure Databricks

Set general and advanced options for advanced properties of the Azure Databricks Create Cluster task.

General Options

The following table describes the general options that you can set for a Databricks cluster:

Property	Description
Cluster Name	Name of the cluster to create.
Databricks Runtime Version	The Databricks version to run on the cluster. Default is the latest supported version. To manually enter a version, select *(Assign to Task Input)*. The Developer tool creates a Create Cluster Task Input called *Databricks Runtime Version*. Set the value of the input to the Databricks version tag string. For example, set the value to `5.5.x-scala2.11` to use Databricks version 5.5.
Python Version	The Python version to run if you include a Python transformation. Default is the latest supported version.
Warm Pool	Check this option to enable creation of the Databricks cluster using a warm pool. Default is unchecked (false).

Property	Description
Pool ID	Value of the DatabricksInstancePoolId property that identifies the warm pool resource. To retrieve the value of this property, go to the warm pool in the Databricks workspace and browse to **Configuration** > **Tags**. Required when the Warm Pool option is checked.
Driver Type	The type of node that you want to use for the driver node. Default is the worker type ID.
Worker Type	The type of node that you want to use for the worker node.
Workers	The number of worker nodes to create for the cluster. If you configure the cluster to scale automatically, this property is ignored. Default is 1.
Autoscale	Automatically scales the number of worker nodes based on workload.
Min Workers	The minimum number of worker nodes to use when the cluster is configured to scale automatically. Default is 0.
Max Workers	The maximum number of worker nodes to use when the cluster is configured to scale automatically. Default is 1.

Advanced Options

Configure advanced options such as environment variables and automatic termination.

The following table describes the advanced options that you can set for an Azure Databricks cluster:

Property	Description
Auto Termination	Enables automatic termination of the cluster.
Auto Termination Time	Terminates the cluster after it is inactive for the specified number of minutes. Enter a value between 10 and 10,000. If you do not configure this, or if you set to 0, the cluster will not automatically terminate.
Cluster Log Conf	The location to deliver logs for long-term storage. If configured, the Databricks Spark engine will deliver the logs every five minutes. Provide the path to DBFS.
Init Scripts	The location where you store init scripts. You can enter multiple destinations. The scripts are run sequentially in the order that you configure them. If you need to install additional Python libraries, specify the init script file location in this property. Use the following format: `dbfs:/<path to init script>,dbfs:/<path to init script>`

Property	Description
Cluster Tags	Labels that you can assign to resources for tracking purposes. Enter key-value pairs in the following format: <key1>=<value1>,<key2>=<value2>. You can also provide a path to a local file that contains the key-value pairs. Use the following format: `file:\\<file path>`
Spark Configurations	Performance configurations for the Databricks Spark engine. Enter key-value pairs in the following format: key1='value1' key2='value2'. You can also provide a path to a file that contains the key-value pairs.
Environment Variables	Environment variables that you can configure for the Databricks Spark engine. Enter key-value pairs in the following format: key1='value1' key2='value2'

Advanced Properties for the Blaze Engine

If you want to use the Blaze engine to run mappings on the cloud platform cluster, you must set cluster configuration properties in the Software Setting property of the Create Cluster task.

Configure the Create Cluster task to set configuration properties in *-site.xml files on the cluster. Hadoop clusters run based on these settings.

Use a text file to specify *-site.xml file settings, and specify the path to the file in the Software Settings property.

The following text shows sample configuration of the Software Settings property file:

```
[
  {
     "Classification":"yarn-site",
     "Properties":{
     "yarn.scheduler.minimum-allocation-mb":"250",
     "yarn.scheduler.maximum-allocation-mb":"8192",
     "yarn.nodemanager.resource.memory-mb":"16000",
     "yarn.nodemanager.resource.cpu-vcores":"12"
   }
  },
  {
    "Classification":"core-site",
    "Properties":{
    "hadoop.proxyuser.<DIS/OSPUSER>.groups":"<group names>",
    "hadoop.proxyuser.<DIS/OSPUSER>.hosts":"*"
  }
 }
]
```

yarn-site

yarn.scheduler.minimum-allocation-mb

The minimum RAM available for each container. Required for Blaze engine resource allocation.

yarn.scheduler.maximum-allocation-mb

The maximum RAM available for each container. Required for Blaze engine resource allocation.

yarn.nodemanager.resource.memory-mb

The maximum RAM available for each container. Set the maximum memory on the cluster to increase resource memory available to the Blaze engine.

yarn.nodemanager.resource.cpu-vcores

The number of virtual cores for each container. Required for Blaze engine resource allocation.

core-site

hadoop.proxyuser.<proxy user>.groups

Defines the groups that the proxy user account can impersonate. On a secure cluster the <proxy user> is the Service Principal Name that corresponds to the cluster keytab file. On a non-secure cluster, the <proxy user> is the system user that runs the Informatica daemon.

Set to group names of impersonation users separated by commas. If less security is preferred, use the wildcard " * " to allow impersonation from any group.

hadoop.proxyuser.<user>.hosts

Defines the host machines from which impersonated connections are allowed. On a secure cluster the <proxy user> is the Service Principal Name that corresponds to the cluster keytab file. On a non-secure cluster, the <proxy user> is the system user that runs the Informatica daemon.

Set the property to " * " to allow impersonation from any host. This is required to run a Blaze mapping on a cloud platform cluster.

Advanced Properties for a Hive Metastore Database

If you want to use a relational database on the cloud platform as the Hive metastore database for the cluster, you must set cluster configuration properties in the Software Setting property of the Create Cluster task.

Configure the Create Cluster task to set configuration properties in the hive-site.xml configuration file on the cluster. Use a text file to specify hive-site settings, and specify the path to the file in the Software Settings property.

Create the following properties in the Software Settings property file:

javax.jdo.option.ConnectionURL

JDBC connection string for the data store.

javax.jdo.option.ConnectionDriverName

JDBC driver class name for the data store. Specify a JDBC driver that is compatible with the cloud platform.

javax.jdo.option.ConnectionUserName

User name to use to connect to the database.

javax.jdo.option.ConnectionPassword

Password for the database user account.

The following text shows sample configuration of the Software Settings property file:

```
{
      "Classification":"hive-site",
      "Properties":{
         "javax.jdo.option.ConnectionURL":"jdbc:mysql:\/\/<RDS_HOST>:<PORT>\/<USER_SCHEMA>?
createDatabaseIfNotExist=true",
         "javax.jdo.option.ConnectionDriverName":"<JDBC driver name>",
         "javax.jdo.option.ConnectionUserName":"<USER>",
         "javax.jdo.option.ConnectionPassword":"<USER>"
      }
   }
```

For example,

```
{
      "Classification":"hive-site",
      "Properties":{
         "javax.jdo.option.ConnectionURL":"jdbc:mysql:\/\/<host name>:<port number>\/hive?
createDatabaseIfNotExist=true",
         "javax.jdo.option.ConnectionDriverName":"org.mariadb.jdbc.Driver",
         "javax.jdo.option.ConnectionUserName":"hive",
```

```
            "javax.jdo.option.ConnectionPassword":"hive"
        }
    }
```

infacmd ccps Command Reference

Use infacmd ccps commands to administer cloud platform clusters.

deleteClusters

Deletes clusters created by the cluster workflow from the cloud platform.

The infacmd ccps deleteClusters command uses the following syntax:

```
deleteClusters
<-DomainName|-dn> domain_name
<-ServiceName|-sn> service_name
<-UserName|-un> user_name
<-Password|-pd> password
<-CloudProvisioningConfigurationID|-cpcid> cloud_provisioning_configuration_id
<-ClusterIDs|-cids> cluster_ids
[<-SecurityDomain|-sdn> security_domain]
[<-ResilienceTimeout|-re> timeout_period_in_seconds]
```

Note: When you use this command to delete clusters on the Azure cloud platform, the process blocks any other command through the command shell until the Azure cloud platform completes the process to release cluster resources. This process could take several minutes. If you try to kill the command using CTRL-C, and then re-run the command, the same time delay and block apply.

The following table describes infacmd ccps deleteClusters options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-CloudProvisioningConfigurationID -cpcid	cloud_provisioning_configuration_id	Required. ID of the cloud provisioning configuration.

Option	Argument	Description
-ClusterIDs -cids	cluster_ids	Required. Comma-separated list of clusters to delete. The cluster ID is the same as the cluster ID listed on the cloud platform site.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. If you omit this option, infacmd uses the timeout value specified in the INFA_CLIENT_RESILIENCE_TIMEOUT environment variable. If no value is specified in the environment variable, the default of 180 seconds is used.
-ConfigurationName -cn	Name of the cluster configuration	Required. Name of the cluster configuration on the domain. Values are not case sensitive.
-DeleteConnections -dc	delete_associated_connection	Optional. Deletes the connections that the cluster configuration created. Use one of the following values: - TRUE - FALSE Default is FALSE.

listClusters

Lists clusters that the cluster workflow creates and that exist on the cloud platform.

The infacmd ccps listClusters command uses the following syntax:

```
listClusters
<-DomainName|-dn> domain_name
<-ServiceName|-sn> service_name
<-UserName|-un> user_name
<-Password|-pd> password
<-CloudProvisioningConfigurationID|-cpcid> cloud_provisioning_configuration_id
[<-SecurityDomain|-sdn> security_domain]
[<-ResilienceTimeout|-re> timeout_period_in_seconds]
```

The following table describes infacmd ccps listClusters options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-CloudProvisioningConfigurationID -cpcid	cloud_provisioning_configuration_id	Required. ID of the cloud provisioning configuration.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. If you omit this option, infacmd uses the timeout value specified in the INFA_CLIENT_RESILIENCE_TIMEOUT environment variable. If no value is specified in the environment variable, the default of 180 seconds is used.

updateADLSCertificate

Updates the Azure Data Lake Service Principal certificate path in a cloud provisioning configuration.

The infacmd ccps updateADLSCertificate command uses the following syntax:

```
updateADLSCertificate
    <-DomainName|-dn> domain_name
    <-UserName|-un> user_name
```

```
<-Password|-pd> password
<-CloudProvisioningConfigurationID|-cpcid> cloud_provisioning_configuration_id
<-CertificateFilePath|-certPath> certificate_file_path
[<-SecurityDomain|-sdn> security_domain]
[<-ResilienceTimeout|-re> timeout_period_in_seconds]
```

The following table describes infacmd ccps updateADLSCertificate options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-CloudProvisioningConfigurationID -cpcid	cloud_provisioning_configuration_id	Required. ID of the cloud provisioning configuration to update with the certificate file path.
-CertificateFilePath -certPath	certificate_file_path	Required. Path to the ADLS Service Principal certificate on the Data Integration Service machine.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. If you omit this option, infacmd uses the timeout value specified in the INFA_CLIENT_RESILIENCE_TIMEOUT environment variable. If no value is specified in the environment variable, the default of 180 seconds is used.

Author

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